

ACCOUNTING

AUDIT PROGRAM #10-00-00

The release of this audit program to parties external to IBM requires the written approval of the Chief Auditor.

This program is intended as a guide for assessing risk management in this process through the performance of an audit or a similar engagement. Risk management is the process to identify, manage, and control potential events in order to provide reasonable assurance related to the achievement of IBM's business and control objectives. IBM's control objectives fall into five broad categories:

1. compliance with external laws and regulations, as well as IBM policies and procedures
2. accomplishment of business objectives
3. reliable information for financial reporting and decision making
4. efficient operations, and
5. safeguarding both intellectual and physical assets.

The control assessment questions are intended to address process objectives from the categories above and the risks to achieving those objectives. Objectives and risks in the actual process under review may differ based on organizational, operational, regulatory, and economic factors. As such, individuals performing the engagement must formulate a test strategy based on risks relative to the actual process under review, regardless of whether the risks are identified within this program.

DATE CHANGED	SECTION #	SUBTITLE IN SECTION	CHANGED BY (NAME)	COMMENTS
December 2025	PLANNING	Audit Program Risk Alignment	Nicolas Garcia Paura	Removed "cybersecurity" from defined risk area of General Ledger Close
September 2025	PLANNING	Audit Program Risk Alignment	CA&AS Ops	Changes to audit risk levels from recent revalidation
February 2025	PLANNING: Data Analysis	All	Nicolas Garcia Paura	Page 5, including S4HANA. Page 7, link update. Page 10, reference information updated with new links.
December 2023	Control Assessment Questions / Key Control	3. Revenue Recognition/Cost Matching	Adriana Lippi	Added optional test: Labor and/or employee concur expense costs
February 2023	All	All	CA&AS	Converted from Notes into a Word document and into a Box folder

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PROCESS AND OBJECTIVE

The general ledger process encompasses the posting of financial transactions from auxiliary or subsidiary ledgers to the general ledger accounts and the creation of accounting transactions based on internal ledger tables (e.g., periodization entries).

For purposes of internal and external reporting, operating decisions, and asset accountability, it is essential that these transactions are executed in conformity with generally accepted accounting principles (GAAP) and within a sound system of internal controls.

This includes the auditing of US GAAP accounting process and the consolidations at the Market, Geography, and Worldwide levels. Also, to include Infrastructure, Consulting, and Financing as reporting segments, and all other externally reported results.

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PLANNING: Audit Program Risk Alignment

Accounting Consolidation

Defined Risk Area	Audit Risk Level	Key Control Question	Test Number
Spreadsheets	High	Processing Integrity	PI007.P
Consolidation Logic	High	Processing Integrity	PI004.P
Accounting Information Warehouse (AIW) Segmentation	High	Processing Integrity	PI004.P, PI005.P
Cash Flow	High	Processing Integrity	PI002.P, PI004.P
Intercompany Float and Eliminations	Medium	Processing Integrity	PI004.P
Year-End Tie Outs	Medium	Processing Integrity	PI004.P
Consolidation Entries	Medium	Processing Integrity	PI004.P
AIW Allocation	Low	Processing Integrity	PI004.P
Force Balancing	Low	Processing Integrity	PI004.P
Miscodes	Low	Processing Integrity	PI003.P

General Ledger Close

Defined Risk Area	Audit Risk Level	Key Control Question	Test Number
Contract Review	High	Revenue Recognition/Cost Matching	RR001.P, RR004.P
Spreadsheets	High	Processing Integrity	PI007.P
Account Reconciliations	Medium	Balance Sheet Account Reconciliations	BS001.P, BS002.P
Ledger Entries, Accruals, and Deferrals	Medium	Processing Integrity Revenue Recognition/Cost Matching	PI002.P RR002.P
Access Management / Separation of Duties / Data Protection	Low	Application and Data Access Authorization (Key Control Question 2, Access Management - IT Enterprise Applications Program 43-10-00) Data Privacy (Key Control Question 4, Data Integrity - IT Enterprise Applications Program 43-10-00)	AM001.P DI004.P
Cost Revenue Match	Low	Revenue Recognition/Cost Matching	RR001.P, RR002.P

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PLANNING: Data Analysis

Data analysis should be a part of every engagement where data is available in databases or data warehouses that can be accessed through query programs. This concept enhances the risk-based auditing approach, makes the audit process more efficient, and increases the productivity of the audit team. When the audit team does not have the skills required to perform these tasks, assistance should be sought from the Technology team.

All data used for analysis must be reconciled to an independent source so that the auditor has confidence that the universe is complete.

The [Data Analysis engagement procedures in the Audit Manual](#) (link internal to CA&AS members only) provide further suggestions.

This table shows what data can be reviewed and analysis techniques.

Data to be Analyzed	Analysis to be Performed	Possible Anomalies	Concern
Manual Journal Entries Request Detailed ledger extract for all manual entries (by system and by accountants, when possible, utilize the IT team for more independent data)	- Look at the dollar amount of the entries (above or below clip levels) - The type/description of entry being booked - The accounts the entry is affecting - The period of the transactions (in regard to month/quarter close) - Request unrestricted reader access to the Global Ledger User Interface (GLUI) and SAP/S4Hana (iERP) to independently review back-up documentation	- Reoccurring amounts just below clip levels - Description of entry not correlating with the accounts being affected - Irrelevant/unnecessary ledger accounts being utilized - Entries without description or with vague descriptions "Corrective Entries" (journal booked then reversed) - Insufficient back-up for journal entries	- Immaterial journals being processed without proper approval - Duplicated journals booked for same accruals or miscode corrections - Journals booked without being authorized or having a valid business justification - Entries were booked using incorrect ledger accounts - Unsupported amounts being journalized (trying to cover up fraud, inappropriate activity) - Entries booked in the wrong time period

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PLANNING: Data Analysis (Continued)

Data to be Analyzed	Analysis to be Performed	Possible Anomalies	Concern
Balance Sheet Account Reconciliations Request a list of all balance sheet accounts reconciled in and out of Blackline Request Blackline access if applicable	- Perform analysis based on content variance, aging variance, and expected account balances - Same reconciliation from month to month for trend analysis - Unresolved and large aged variances - Large swings from month to month - Zero variances with significant content balances	- Expected account balances are not supported by valid back-up documentation - Variance items are not properly aged (month to month) - Same reoccurring entries throughout the months for the year	- Unidentified activity could be occurring in accounts - Variance items not aged appropriately could lead to misappropriate reporting - Lack of sufficient documentation to explain and resolve variances could lead to financial misstatements

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PLANNING: Reference Material

The Worldwide Accounting Practices (WWAAP)

IBM Accounting Practices establish accounting principles to be used by all organizations. Practices are based on U.S. GAAP as established by the duly authorized standard setting bodies: the Financial Accounting Standards Board (FASB), the American Institute of Certified Public Accountants (AICPA), and the Securities and Exchange Commission (SEC). IBM financial statements must be prepared based on the broad fundamental rules, policies, and procedures stated in the Practices.

Link: <https://w3.ibm.com/w3publisher/accounting-policy-financial-reporting/accounting-practices-interpretations>

The Worldwide Accounting Corporate Accounting Instructions (WWACAI)

IBM Accounting Instructions provide detailed guidance that must be followed by all organizations to ensure worldwide consistency in the implementation of IBM Accounting Practices.

Link: <https://w3.ibm.com/w3publisher/accounting-policy-financial-reporting/accounting-instructions>

Worldwide Chart of Accounts (WWCOA)

The WWCOA establishes and defines the required account structure that must be used in the General Ledger of the Corporation. This structure facilitates the consolidation of worldwide finance data and provides a consistent basis in the preparation of financial reports for external reporting and internal measurements.

Access via the following link: <https://gart.us1a.cirrus.ibm.com/wwcoa/summary>

Worldwide Accounting Controls

Provides reference material, link to corporate guidelines, accounting instructions and practices.

Link: <https://w3.ibm.com/w3publisher/worldwide-accounting-controls>

Corporate Instructions

The IBM Corporation's primary methods for documenting its policies, delegations, and instructions to IBM managers and employees are contained in the Corporate Directives. These authorities consist of policy letters, organization and staff letters, and corporate instructions. These are for example:

FIN101: Control of disbursements

FIN - HR 162: Financial and Human Resources Applications

FIN166: Risk evaluation documentation - Corporate Directive Deviations

FIN167: Vendor sourced labor services

FIN 182 Management Self-Assessment of Control Appendix C – Separation of Duties

<https://w3.ibm.com/w3publisher/financebc/ibm-framework/ci-fin-182-appendix-c>

Global Ledger: Functional Guidance, Data Standards, Strategy and Transformation

This page is a repository for global standards, guidance to the General Ledger and ledger processes, and information that supports the CLS-FDW processes.

Link: <https://w3.ibm.com/w3publisher/americas-ledger-operations/cls-fdw>

Regional Accounting Centers

The website provides frequently used information for the Regional Accounting Centers in Buenos Aires, Argentina; Bratislava, Slovakia; Budapest, Hungary; Kuala Lumpur, Malaysia; and WW Accounting Controls in Armonk, New York.

Link: <https://w3.ibm.com/w3publisher/act-count-think/locations>

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PLANNING: Fraud Template

Fraud is any intentional act or omission designed to deceive others and resulting in the victim suffering a loss and/or the perpetrator achieving a gain. Fraud can be categorized into fraudulent financial reporting, misappropriation of assets, and improper or unauthorized expenditures.

Description of Fraud Risk	Typical Controls that should Detect the Fraud Risk	Testing Approach
Authorization, Recording, Custody, and Reconciliation roles	Separation of Duties Assessment	• Review separation of duties assessment to validate: • The end-to-end process is reviewed annually by management • Assessments are kept current and are updated when organization, staffing assignments, process or system changes occur that significantly impact the SOD assessment • The approach to separating duties, whether by process design, system controls, organizational structure, or other means is established and maintained on an on-going basis • Identified conflicts are assessed for SOD risk and mitigating controls are implemented effectively to separate conflicting tasks
Falsifying Financial Statements	Review and approval of the financial reports coming from different accounting organizations Key Controls Over Financial Reporting	• Select a sample of journal entries (manual adjustments, miscodes, system generated entries, etc.) to review the sufficient documentation and appropriate levels of approvals • Select a sample of key (cash, cash equivalents, etc.) accounts and review the reconciliations and their content, backup, and necessary approvals • Review a sample of contracts for revenue and cost recognition and validate that with appropriate regulatory guidance

Note: The probability and impact will vary depending on the organization/business unit that is being audited.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Balance Sheet Account Reconciliations

Objective:

Are all balance sheet accounts reconciled per the approved frequency (i.e., monthly, quarterly, annually, etc.) with valid supporting documentation and the appropriate management's review and approval?

Risk/Exposure:

Reconciliations performed correctly and on time are the primary control to identify any differences between the general ledger and the systems which feed it. Failure to perform reconciliations consistently indicates a lack of management focus on essential financial controls and could lead to a misstatement of the company's financial position.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Balance Sheet Account Reconciliations

Primary Test: Account Reconciliation Performance (BS001.P)

Objective:

To ensure that account reconciliations are performed on a regular basis; and that reconciliation activity is supported by a management system which ensures control and performance of reconciliations for all accounts.

Approach:

Request the ledger accounts that this audit unit owns and reconciles.

Ensure that the account group were valid, and the description of the reconciliations were current.

Reconciliations have to be performed for all balance sheet accounts that are loaded to Blackline and do not auto-certify per the Blackline criteria. If an AIW/GLUI-C-only posting is made to a ledger account that has a zero balance with no other current month activity, no hard copy reconciliation is required to be prepared for that particular month. The review and approval of the posting as part of the regular posting process is sufficient for documenting the validity of the activity. If the AIW/GLUI-C-only posting is then ledgerized the next month, the activity will flow naturally to Blackline and will require a reconciliation per the standards set forth in the guidance.

Ensure that selected reconciliations were supported with sufficient back up documentation and proper approvals and in line with requirement stated in AI CTRL 12-005. In the selected samples, make sure the entries or the activities recorded in the reconciliations were classified in the appropriate categories (content/variance aging), tracked and aged until they were cleared. When the content items exceed the expected clearing date, they should be reclassified to variance items and action should be taken when applicable. For any variance items aged over 180 days and over the \$20k clip level, action plans should be put in place and tracked to make sure they will be cleared and prioritized.

If there were any items in the reconciliations that could not be supported or identified, they should be called out as unreconciled items and should be reported properly and timely to the management. If there were any items that were written off, review the justification and management approvals.

Document evidence of compliance testing and/or a quarterly review with accounting management's approval to validate the reconciliations are performed in accordance with the accounting guidelines.

References:

AI 14-003 Write Off Guidance

AI 12-005 Standard for Reconciliations of Balance Sheet Accounts

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Balance Sheet Account Reconciliations

Primary Test: Account Reconciliation Content (BS002.P)

Objective:

To ensure that account balances are accurately reconciled with supporting analyses and documentation.

Approach:

Select a sample of asset and liability account reconciliations in order to test the effectiveness of the reconciled activities. If applicable, include as a minimum:

- Cash and Bank
- Inventories
- Investments
- Reserves/provisions
- Accruals and prepayments
- Clearing and suspense accounts
- Intercompany transfer accounts
- Control accounts (e.g., Accounts Receivable and Accounts Payable)
- Control accounts for items awaiting recording to sub-ledger (e.g., in-transit account for inventory)
- Unbilled accounts receivable
- Cash receipt control
- Business Optimization Project (BOP) memorandum accounts

Sample selection within the above categories should be supported by analysis of trends in accounts balances, including:

- Unusual trend
- Significant balance changes - big deltas
- High volume accounts, level of amounts
- Unexpected credits/debits
- New/closed balances
- Account balance contradictory to account type
- Non-flat trends for capital accounts
- Sensitive accounts (write-offs, accruals, etc.)

Review the reconciliations and their content and determine the following:

- Is the period (accounting month and year) being reconciled stated on the balance sheet reconciliation?

Was the account reconciliation prepared on a timely basis? If the reconciliation is not required on a monthly basis, make sure the guidelines are adhered to for quarterly and annual reconciliations.

Was the reconciliation completed by due date as defined by AI CTRL 12-005?

- Monthly – by the last day of the calendar month following the accounting period requiring a reconciliation (for example, no later than April 30 for the March accounting period)
- Quarterly – Key reconciliations same as Monthly (for example, no later than April 30th when reconciling 1Q activity). Non-Key reconciliations by end of the 2nd month following the quarter (for example: no later than May 31 for a 1Q Non-Key Quarterly recon)
- Annually – same as Monthly with it being completed on the last day of the month following the ending accounting period in the reconciliation (for example, no later than Jan 31 for reconciliation period ending in December)

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Balance Sheet Account Reconciliations

Primary Test: Account Reconciliation Content (BS002.P)

(Continued)

Was there any exception, if any, of the reconciliations were not performed or approved by the due date?

Was the approver properly authorized and delegated?

Was the operational balance verified with independent source document, e.g., bank statements to cash accounts, deferred revenue to the corresponding amortization schedule, PDA accounts to the applicable calculation, invoices, copies of checks and schedules/spreadsheets, etc.

Was the current ledger balance tie to the sub-ledger accounts or sub-feeders?

Were these variances being aged properly in the correct aging categories?

Was there evidence of account manipulation in the reconciling items and adjustments, e.g., Journal entries being processed to alter the entry date of the reconciling items to improve measurement results? Journal entries being processed to write-off the reconciling items without the proper level of management approval?

Were aged variances over 180 days properly reported in the quarterly CHQ Aged Activity Scorecard?

Reference:

Standard for Reconciliation of Balance Sheet Accounts

<https://w3.ibm.com/w3publisher/worldwide-accounting-controls/accounting-instructions>

AI 14-003 Write Off Guidance

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Processing Integrity

Objective:

Is there adequate separation of duties between the individuals reconciling accounts; those preparing and approving journal entries; those updating ledger processing control tables; and performing activities in the processes which feed the accounts (e.g., handling customer accounts or approving payments to vendors, updating system tables).

Are there adequate controls over journal entries; feeder system input and miscode corrections and ledger processing table update?

Risk/Exposure:

Failure to maintain adequate separation of duties could allow individuals to manipulate accounting records and mask any irregularities from other systems which feed the ledger.

Ineffective journal entry control may enable employees to conceal irregular entries or clear reconciling items from account analysis by writing them off or otherwise disguising the real nature of the activity.

Lack of control over feeder system input, miscode correction and ledger processing control tables can allow reported accounts to become out of line with supporting ledgers.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Processing Integrity

Primary Test: Separation of Duties (PI001.P)

Objective:

Ensure that a separation of duties assessment is performed and approved by the process owner, Center Leader, or delegate annually, or more frequently as necessary. Validate that identified separation of duties conflicts are eliminated, mitigated, or accepted based on risk severity. Ensure that conflicting tasks which should be separated, but are not, have effective secondary controls in place.

Approach:

Request an assessment of tasks executed by roles within Accounting. If no assessment exists, determine whether management has alternative methods of identifying potential separation of duties conflicts.

Review the SOD Assessment to validate the following:

- Was the end-to-end process reviewed at least annually and at the appropriate level? Were the results documented and approved by the process owner, Center Leader, or delegate?
- Were tasks assessed to identify conflicts in the SOD Matrix and were identified conflicts explained?
- If secondary controls are needed, were they identified and would they mitigate the conflict effectively?
- Were conflict mitigation steps executed as outlined on the SOD assessment?

Review the responsibilities assigned in the areas of accounting for possible separation of duties exposures. Determine if:

- A SOD analysis exists for each role to identify any potential conflicts and implemented secondary controls are appropriate.
- Revised departmental process and revised departmental mission are timely reflected in the SOD analysis.
- The requirement is for the SOD to be updated annually.
- Assess if any role changes that aren't included in the SOD, cause any conflicts.

Individuals who can perform activities which combine custody (possession or guardianship over assets), recording (creation of permanent transaction record), authorization (transaction or document approval) and verification (verification of the accuracy and completeness of transactions) are identified as potential conflicts and secondary controls are implemented to mitigate the respective exposure.

Include in the review any update authority to systems, system tables or data. No individual should have the capability to approve and enter the same journal entry. Any individual with capabilities which would allow them to circumvent preventive and detective controls in the processes should be identified and included in test samples.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Processing Integrity

Primary Test: Manual Ledger Entry Controls (PI002.P)

Objective:

To verify that there is a management process to ensure that only properly authorized manual entries are posted to the general ledger, and to determine whether this process is working effectively.

Approach:

Note that manual ledger entries will typically be made using journals, but the auditor should ensure that all sources of manual ledger entries (e.g., feeder systems with general ledger functionality) are identified for inclusion in testing.

Select a sample of accounting journal entries to determine:

- Did the journal voucher contain correct account coding, month of entry, voucher number, date prepared, the preparer's signature, and the approval authority signature?
- Were the entries properly reviewed and approved by the appropriate level of management?
- Were entries supported with sufficient and appropriate documentation?
- Were journal vouchers processed in a timely manner?
- Was the entry in accordance with the generally accepted accounting principles and IBM practices?
- Were write-offs supported with business justifications and proper approvals?
- Were there clear audit trails for ledger entries, including information regarding the originators and approvers of the entries?
- Was there original entry to be supported for the reversal entries?

Review allocation tables to ensure that the table was setup correctly with the appropriate percentages and the accounting codes, and that the tables were updated periodically to capture the changes.

Reference: Preparation and Approval of Manual Ledger Entries

<https://w3.ibm.com/w3publisher/worldwide-accounting-controls/accounting-instructions>

AI 14-003 Write Off Guidance

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Processing Integrity

Primary Test: Miscodes / Errors (PI003.P)

Objective:

To determine whether there are controls in place to ensure miscodes are properly reviewed and approved for processing into the ledger. There are certain miscodes that are required to be corrected and others that are allowed to be booked to a default account.

The following miscodes must be corrected to the account they were intended to charge:

1. All balance sheet and revenue miscodes; regardless of the clip
2. All product or receiving country business miscodes; regardless of clip
All remaining miscodes (account and business) over \$10K USD (at Actual rates). See reference AI CTRL 14-002 Financial Data Quality (Miscodes) for details of change log.

Once the above three types of miscodes are corrected, the only miscodes remaining would be cost and/or expense miscode under \$10K. These miscodes are to be corrected to the default account major minor 920 9999.

The activity is corrected to the 920-9999 default account is to remain there unless the miscode is related to balance sheet activity or notification is received requesting corrective action due to a material miscode having been corrected to the default account in error. It is not acceptable to make entries out of the default account for any other reason.

The clip level should be applied at Actual rates using the previous month currency exchange rate as the current month exchange rate is not known until critical workdays. It is expected that Accountants should be correcting miscodes throughout the month and not be waiting until the end of the close to correct all miscodes. The default account is only to be used by the Accounting organization. Individuals outside of Accounting who have miscode correction responsibilities are NOT to use the default account.

There are rare situations where use of a clip level is prohibited for legal and contractual reasons with the US federal government. These organizations (e.g., GBS Federal and IBM Research) therefore should investigate and correct all miscodes in order to stay in compliance with contractual requirements. The miscodes resulting from these areas; however, will still be included in any communication with feeders in order to continue to drive efficiencies.

Approach:

Are there any miscodes that are sitting in the default account that should have been cleared? If so, is there a business justification as to why they have not been cleared?

Reference: AI CTRL 14-002 Financial Data Quality (Miscodes)

<https://w3.ibm.com/w3publisher/worldwide-accounting-controls/accounting-instructions>

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Processing Integrity

Primary Test: Worldwide Consolidation Accounting (PI004.P)

Objective:

Verify that ledger data was recorded and reviewed for accuracy, validity, and completeness during worldwide consolidation.

Approach:

Review the Worldwide Consolidation Accounting (WCA) website to understand the requirements for Consolidation GLUI entries and key processes. Link: <https://w3.ibm.com/w3publisher/wca>

Identify the management controls applied to ensure the accuracy, validity and completeness of ledger data transferred to AIW/WWIBM.

Obtain a listing of the respective control reports (tie outs) which are created to validate the data transfer from AIW (Accounting Information Warehouse) to AIW/WWIBM. Sample reports and review them for completeness and timeliness. Verify if variances were supported with coherent documentation and explanation, if they were timely reviewed by management and corrective actions taken, as appropriate.

Controls to ensure eliminations are performed correctly. Verify if inter-company floats (in transit items such as charges or credits issued by one company to another, but which have not been recorded by the receiving company) are analyzed and reviewed by management. Unusual trends have to be examined and explained. Request a listing of all respective spreadsheet (e.g., IS&S / Inter-Company Shipment and Services). Select a sample of spreadsheets ensuring coverage of each major process and verify if variances were identified and supported with coherent documentation and explanation. Verify if they were timely reviewed by management and corrective actions were taken as appropriate. In addition, sample data on the reports and compare to the system to ensure correctness.

Controls to ensure key accounts and transactions have been reviewed to ensure accuracy of AIW/WWIBM data. Review with auditee the controls implemented to gain knowledge of the control reports / procedures / queries performed during the closing to ensure the accuracy of WWIBM. These controls should include checks which determine if the inter-company accounts have a zero balance, if all standard voucher entries were processed, etc. Sample key control testing from some months and verify if the controls were performed, all issues were identified, and management was aware of these issues. In addition, verify if issues were investigated and corrective action taken as appropriate.

Control and approval process to ensure that all significant data has been analyzed and approved before being reported in the external financial statements. Sample control reports such as Key Control Indicator (KCI) reports, Trend Analysis Reports etc. and verify if they were reviewed and approved by management according to the delegation and if the data in these reports match with the data in AIW/WWIBM.

Controls to ensure the reported results were validated and approved. Various system tie-outs ensure that the data in the financial systems are accurate and completed. Obtain a schedule of all tie-outs performed (AIW to Ledger, AIW to WWIBM, AIW, etc.). Review with auditee the control procedures implemented to gain knowledge of the controls performed including the reports and reconciliation used. Verify if these controls are sufficient to identify mismatches and incorrect data. Sample the respective reports ensuring coverage of each tie out, focus on quarter-ends and determine if the controls were timely performed, reviewed and approved by management, variances were identified, root cause analysis was performed, and corrective actions were taken.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Processing Integrity

Primary Test: Worldwide Consolidation Accounting (PI004.P)

(Continued)

Controls to ensure no additional data can be entered in WWIBM/AIW/GLUI Consolidation after it has been finalized. Review the control procedures implemented to gain knowledge of the controls to ensure no additional data is entered. Verify if these controls are sufficient to prevent data to be entered. Sample the respective control documentation and verify if the controls were performed as required (e.g., systems and applications were closed / frozen). If applicable compare final reported figures with the final approved figures to identify unauthorized variances which might have been added after final management approval.

Controls to ensure new or changes to major, minor, subminor, Account Ownership, WW Divisions, TB Countries, Product Reference data, Customer Set, Currency Rates, and Financial Rules (e.g., Consol_Logic, Brand segmentation, allocations, etc.) are reviewed and tested by WCA and approved by management, prior to implementation. Sample system changes performed and verified if there is sufficient documentation to support the change (requestor form, etc.), all necessary approval were obtained, the change requested and approved was implemented as requested and approved, and the change was tested as appropriate prior to the system change. Verify if business rules are updated as appropriate and validated periodically. Obtain a listing of all business rules changes and verify if these were timely reflected in the business rules stored in the application.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Processing Integrity

Primary Test: Accounting External Reporting (PI005.P)

Objective:

Verify that external reports are reviewed for accuracy and completeness and are approved prior to publishing.

Approach:

Verify if the final external report which was published was reviewed by / with the Review Team, the Disclosure Committee, the Executive Management, and the Audit Committee and approved by the VP of Accounting Practices and External Financial Reporting and External Reporting (Senior External Reporting Program Manager). Sample items from the external report and verify if they tie to the internal figures (ledger / AIW / WWIBM).

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Processing Integrity

Primary Test: Business Entities in which IBM has a controlling interest (PI006.P)

Objective:

Accounting for Business Entities in which IBM has a controlling interest.

Verify that the financial statements are consolidated to present the financial position and results of operations of IBM and those business entities in which it has a controlling interest as if they are one entity.

Approach:

Obtain a list of all IBM's business investments (e.g., subsidiaries, joint ventures). Determine which consolidation method was used (cost method, equity method and line by line consolidation). Verify if the consolidation method used was in accordance with the IBM accounting practices. If an inappropriate consolidation method was used determine the financial impact the use of the wrong method had.

Review the consolidation process and controls for line-by-line consolidations. Determine the controls which ensure that offsetting intercompany accounts, intercompany profits and intercompany dividends are correctly eliminated as well as the parent's company's investment account against the equity account of the investee.

Sample business entities which are consolidated line-by-line. Verify if IBM's investment account was eliminated against the equity account of the investee.

Sample intercompany accounts receivable and accounts payable transactions. Verify if they were eliminated.

Sample transactions where the investee delivered for IBM to an external customer, the investee delivered for IBM internal projects and IBM delivered for the investee. Determine that the intercompany revenue and cost were eliminated.

For all samples where the investee is not wholly owned by IBM verify that the minority interest was recorded in the balance sheet.

For non-US countries determine if the local accounting laws permit capitalization for products which are not allowed to be capitalized under US-GAAP (e.g., SW licenses). Determine if the investee capitalized these products and if the capitalization was eliminated in the IBM ledger.

ACCOUNTING

AUDIT PROGRAM #10-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Processing Integrity

Primary Test: Spreadsheets (PI007.P)

Objective:

To verify that there is a process in place to ensure that spreadsheets were reviewed and approved by management, and the results of calculations were reflected correctly in the ledger. The intent is to provide a framework to ensure there are appropriate controls within the spreadsheet to prevent errors and the spreadsheet is working effectively.

Approach:

Note: Test approach is aimed to test spreadsheets included in the process owner's spreadsheet inventory potentially requiring a Management Checklist or ASCA certification.

Request the spreadsheet inventory from the process owner and evidence of last approval. If the process owner does not have a spreadsheet inventory understand why one is not required or was not performed.

Impact to the sub-process should be considered when determining which spreadsheets to include in the sample selection.

Sampled Spreadsheets

Select a sample of spreadsheets to determine:

1. Has the spreadsheet been reviewed annually to ensure proper controls are in place? If applicable, does the response align with question #1 on the Management Checklist?
2. Is the spreadsheet working as intended? Check the following:
 - Validation key input is for correct period
 - Review/sample key calculations
 - Review check total/reconciliation
 - Review ties out to source documentation
3. If the spreadsheet contains a summary to generate either an entry or chart, does the summary tie back to the detailed calculation(s)?

ASCA Certification

If the sampled spreadsheet is ASCA certified, use the below question:

4. If the spreadsheet requires ASCA certification, is there a valid ASCA certification in place? Assess whether the developer and user is the same person? If they are, then review that the documented secondary controls in place are appropriate and being executed as designed as to mitigate SOD risk.

ACCOUNTING

AUDIT PROGRAM #10-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Processing Integrity

Primary Test: Spreadsheets (PI007.P)

(Continued)

Management Checklist

If the sampled spreadsheet has a management checklist, review the spreadsheet to confirm that:

5. Review management checklist to ensure that it is approved (by manager signature and date).
6. Are the spreadsheet controls and logic documented? Does the response align with question #2 on the management checklist?
7. Is there a documented procedure on how the spreadsheet is used? Does the response align with question #3 on the management checklist?
8. Is there an identifier in the spreadsheet that is updated each time the spreadsheet is executed to reduce the risk of using data or calculations for incorrect periods? Is this documented in the procedures? Does the response align with question # 6 on the management checklist?
9. Is there an automated or manual check in place to confirm manual and automated financial inputs are entered into the spreadsheet correctly? Does the response align with question #4 on the management checklist?
10. Are there steps in place to ensure any updates are intentional and to catch unintentional updates? Does the response align with question #10 on the management checklist?
11. Is there a procedure that documents when a new control review of a spreadsheet is needed? Does the response align with question #8 on the management checklist?

Reference: AI-CTRL 12-012 Accounting Spreadsheet Instruction

<https://w3.ibm.com/w3publisher/worldwide-accounting-controls/accounting-instructions>

ACCOUNTING

AUDIT PROGRAM #10-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Revenue Recognition/Cost Matching

Objective:

- Is there an effective management review process to ensure that all revenue is recognized in the financial statements in accordance with generally accepted accounting principles, IBM practices and instructions?
- Is there an effective process for matching cost to revenue to ensure the integrity of cost/revenue recording and gross profit reporting?
- Is there any evidence of improper practices in any of the shipments or other activity for which revenue has been recognized?

Risk/Exposure:

Inaccurate cost and revenue reporting could have a significant impact upon the integrity of the IBM financial statements.

Where revenue recognition is not triggered by system interfaces, (for example services revenue and systems integration contracts) there is a risk that achievement of measurement targets or commission incentives may encourage misstatement of revenue.

ACCOUNTING

AUDIT PROGRAM #10-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Revenue Recognition/Cost Matching

Primary Test: Revenue Recognition and Cost/Revenue Matching (RR001.P)

Objective:

To ensure that an effective process exists to ensure revenue and cost were recognized in accordance with generally accepted accounting principles and IBM practices.

AP REV: Revenue Recognition

<https://w3.ibm.com/w3publisher/accounting-policy-financial-reporting/accounting-practices-interpretations>

Approach:

It is key that the revenue recognition rules relevant to the contract and products or services being reviewed are understood.

Perform a process control interview with the appropriate accounting management and verify the current revenue recognition methodology.

Note these principles apply to revenue recognized prior to January 2018:

There are basic principles for revenue recognition which always have to be met. These are:

- Persuasive evidence of an arrangement between IBM and the customer exists (firm order)
- Delivery to the customer has occurred or services have been rendered (The customer has taken title and risk of loss has passed to the customer such that the customer has assumed the risk and rewards of ownership.)
- The arrangement fee is fixed or determinable
- Collection of the arrangement fee is reasonably assured
- Cost of revenue has to be recorded accordingly

Note these principles apply to revenue recognized after January 2018:

IBM shall recognize revenue to depict the transfer of promised goods or services to customers in an amount that reflects the consideration to which IBM expects to be entitled in exchange for those goods or services. To achieve that principle, IBM shall apply the following steps:

- Step 1: Identify the contract with a customer
- Step 2: Identify the performance obligations in the contract
- Step 3: Determine the transaction price
- Step 4: Allocate the transaction price to the performance obligations in the contract
- Step 5: Recognize revenue when or as IBM satisfies a performance obligation

For each major revenue reporting category (hardware; software; services) and significant contract types identify the management controls applied to ensure the integrity of revenue recognition and cost matching (e.g., Cash must be reasonable assured when revenue is recognized. Determine if adequate controls are in place to ensure that no revenue is recognized for customers in suspense or bankrupt customers where IBM is not sure to receive payment for the deliverables. Check especially the controls for periodized revenue like leasing or maintenance.).

Select a sample of revenue items, ensuring coverage of each major revenue category and contract type. When selecting a sample to test revenue recognition consider quarter-end, year-end activities and inception to date values, items on unbilled balance sheet accounts, aged accounts receivables, revenue recognized for customer in suspense / bankrupt customers, claims, credit notes, transactions where revenue was reversed in the following months and canceled transactions. They might give you indications about revenue overstatements.

ACCOUNTING

AUDIT PROGRAM #10-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Revenue Recognition/Cost Matching

Primary Test: Revenue Recognition and Cost/Revenue Matching (RR001.P)

(Continued)

For the contract review assessment, have list of all firm order numbers for HW/SW, Services Project Based Services (PBS) work numbers/projects active during the audit period and for Services Maintenance contracts active during the audit period available for review when the team arrives on site. This universe is to include 100% of the TCV / firm orders with start date in audit period, and 100% of the invoices issued/billed during the audit period regardless of who reviews the contracts for revenue and cost recognition.

- This includes all contract modifications (e.g., amendments, change notes, etc.).
- These can be hardcopy, digital format, access to the systems these reside in, or a combination.
- Demonstrate that this captures 100% of the universe.

The "Global Checklist for Conditions Requiring Accounting Review" may also provide sampling guidance. Review the document prior to sampling to ensure that high risk transactions are selected. The checklist is periodically updated to reflect changes in accounting interpretation and must be completed for contracts where there are any additions, deletions, or other modifications made in the standard engagement terms and conditions, base agreements, term agreements, or any other supporting contractual documentation. In addition, also obtain delegation levels for the respective process and sample multiple transactions that fall slightly below clip levels.

Determine if all four basic principles for revenue recognition were met when revenue was recognized. Verify delivery/shipment to actual delivery/shipping documents from labor claiming systems, subcontractor labor documentation, and carrier/transporter and track revenue and the related costs to the ledger. Ensure that you request all documents required for testing (e.g., firm order, proof of delivery, customer credit check).

Also verify the following:

Matching of product cost to revenue by product and revenue category was performed monthly to ensure the correct reporting of gross profits. In addition, verify if all applicable hardware, software, and service costs were charged in the same month when the associated revenue is recognized.

Revenue was not recognized if there were any contingencies that have not been satisfied by the end of the accounting period? These could be pending IBM or customer approval of the agreement or related documents (e.g., by management, legal department, Board of Directors, etc.), execution of another agreement or related documents (e.g., IGF Term Lease Supplement). The order is not considered firm, and revenue must not be recognized until the contingency is resolved or expires.

Revenue was not recognized if there were any unsatisfied customer acceptance provisions or performance guarantees?

Contractual customer acceptance provisions are considered substantive, bargained-for terms of an arrangement and therefore revenue generally should not be recognized until customer acceptance occurs or the acceptance provisions lapse. Note: This is not applicable to contracts accounted for under the percentage of completion (POC) method of accounting, such as certain services contracts.

ACCOUNTING

AUDIT PROGRAM #10-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Revenue Recognition/Cost Matching

Primary Test: Revenue Recognition and Cost/Revenue Matching (RR001.P)

(Continued)

The revenue portion of the delivered product's fee that is withheld by the customer or that is refundable was deferred until the perfunctory undelivered product or service was delivered?

No revenue recognized for all "bill and hold" transactions? Revenue must not be recognized for transactions in which the customer is billed for a product but the product is either not shipped at the time of billing or is physically delivered to an intermediate destination that is other than the customer's business site or another site specified by and under the control of the customer.

No revenue was recognized for products shipped under a demo, loan, or trial program until the customer finally bought the product?

Revenue was deferred until cash collection for arrangements above a certain clip level and with deferred payment arrangements (e.g., balloon payments). Request the valid WW guidelines for details on criteria and clip levels.

For revenue related to resale of other vendor's products and services a gross versus net analysis was performed and the revenue was recorded either gross or net in accordance with the current Accounting Practice.

Deals where "roundtripping" or revenue "swaps" could be suspected were reviewed by the applicable Director of Accounting and it was ensured that IBM avoided roundtripping accounting and transactions were accounted for based upon its economic substance. There are almost an infinite number of scenarios that would qualify as roundtripping or revenue swap. "Roundtripping" or "swapping" revenue generally involves selling something to a vendor that was previously purchased from that vendor or selling different products back and forth that are negotiated simultaneously and, in some cases, can occur from bartering transactions (e.g., Non-Monetary Exchanges).

All I&E exposure (contingent expenses and losses) which exists as a result of current or past events was recorded in the current period based on reasonable estimates, if it is probable that a future confirming event will occur (e.g., Due to unexpected increase of efforts/cost for a service fix price contract the contract will generate a loss at the end of the contract. This loss must be recorded immediately.)

The Estimate of Future Billing Adjustment (FBA) Provision for services time and material contracts, hardware and software was performed in accordance with the respective accounting guidance and the values recorded are justified.

The reported revenue value takes adjustments required by the IBM accounting practices into account (e.g., sales returns, trade-ins).

Determine if there is an effective management system in place to ensure that no revenue is recorded until a product is announced as being generally available. Note: If the customer participates in an Early Support Program for software products or an Early Ship Program for hardware products revenue may be recognized when the requirements outlined in the accounting rules are met.

ACCOUNTING

AUDIT PROGRAM #10-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Revenue Recognition/Cost Matching

Primary Test: Revenue Recognition and Cost/Revenue Matching (RR001.P)

(Continued)

In addition, review the "Global Checklist for Conditions Requiring Accounting Review." All conditions on the checklist should be verified against the sample of transactions. This is a document updated periodically and is required for any contracts where any additions, deletions and/or other modifications made in the standard engagement Ts & Cs, base agreements, term agreements, or any other supporting contractual documentation were made.

The following revenue recognition criteria are product specific and should be considered when testing one of these products categories. For multiple element arrangements see the separate test in the program (RR004.P).

HARDWARE REVENUE:

Verify that revenue was recognized when the customer has taken control and risk of loss has passed to the customer. No revenue is to be recognized until the products are delivered to the customer and all other revenue recognition criteria are met when shipment occurred to an IBM owned site or to a third-party site that is rented, subsidized, or otherwise controlled by IBM. Note: if the customer specified an intermediate site which is also a customer's business site or another site specified by and under the control of the customer for delivery, revenue can be recognized in case all revenue recognition criteria has been met, special attention should be placed on transfer of control and if the customer has the risks and rewards of ownership of the Hardware at this intermediate site.

If specific requirements outlined in the accounting rules are met revenue might be recognized when the hardware is shipped. However, in those instances the fair value of the installation services should be deferred, and the cost of installation should be recorded at the time of installation.

Determine if there is an effective management system in place to identify extended warranty arrangements. Verify that revenue for extended warranties was deferred and recognized ratably over the period in which the services are provided and not recorded upfront upon hardware delivery like for standard warranties.

Determine if there is an effective management system is in place to identify when excess inventory is held by Business Partners such that the distribution channel is overstocked with inventory (i.e., channel-stuffing). Verify that when excess inventory within a distribution channel exists, revenue and cost were reversed as required by the accounting rules.

SOFTWARE REVENUE

The following are the typical main IBM software offerings and the respective revenue recognition rules assuming you can unbundle hardware/software using vendor-specific objective evidence or the software in bundled hardware/software offerings was not incidental to the offering.

Recurring License Charge (RLC)

Revenue from RLC arrangements, including free releases and support, is earned in the month of usage by the customer. For administrative simplicity, billings may be the basis for revenue recognition provided that it results in recognition in the appropriate period, e.g., quarterly.

ACCOUNTING

AUDIT PROGRAM #10-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Revenue Recognition/Cost Matching

Primary Test: Revenue Recognition and Cost/Revenue Matching (RR001.P)

(Continued)

Maintenance Agreements (MA)

Revenue for MA is earned ratably over the period during which the service is provided. For administrative simplicity, billings may be the basis for revenue recognition provided that it results in recognition in the appropriate period, e.g., quarterly.

Subscriptions

Revenue is earned ratably over the period during which the product/subscription is provided

One Time Charge (OTC)

Single License: should be recognized upon delivery of software.

Master Diskettes:

Fixed Fee: the delivery requirement is satisfied when the master diskette is provided to the customer. The fee is fixed when both the price and number of copies are specified, and the contract is not cancelable.

Under these conditions, revenue may be recognized upon delivery.

Per Copy: when the fee is based on the actual number of copies made, revenue should be recognized at a rate per copy when the copies are actually made.

Verify in your testing that software revenue was recorded ratably over a period of time was not recorded like One Time Charge software.

Verify that intercompany software license fee (SLF) and royalties, which are calculated as a percentage of revenue, were recorded at the same time the revenue was recorded.

SERVICES REVENUE

The following are the typical main types of service contracts and the respective revenue recognition rules. When testing services contracts verify if no revenue was recognized for work at risk arrangements. They do not qualify as firm order.

Time and Material Basis

Revenue is earned at the time the service is provided according to reported hours.

Fixed Period

Revenue is recognized straight-line over the period the service is provided or other than straight-line when a pattern of services is identifiable (e.g., outsourcing with increasing level of services)

Revenue for Maintenance Agreements which have a quarterly billing cycle will be recorded at the time of billing.

Unique Product/Services

Revenue is recognized based on a calculation which uplifts inception to date incurred costs, including the cost of OEM hardware and software products, by the contract gross profit percentage based on the latest estimates of total costs for the entire contract (percentage of completion methodology / POC).

ACCOUNTING

AUDIT PROGRAM #10-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Revenue Recognition/Cost Matching

Primary Test: Revenue Recognition and Cost/Revenue Matching (RR001.P)

(Continued)

For POC contracts verify if:

- revenue was recorded at a zero gross profit percentage when there were significant and frequent changes in the estimates of total contract costs, and/or there is significant risk in meeting the customer requirements
- revenue accruals were ceased, and all previously recognized revenue was reversed when the ultimate performance became questionable
- revenue was not accrued unless the billing will occur within six months

System Operation Contracts (Strategic Outsourcing)

Revenue must reflect the extend of actual services delivered in the period. The method to measure the extend of services delivered should be a measure of output stated in the contract as the IBM deliverable. Verify if:

Accrued revenue was sufficiently covered by termination charges

Revenue from transition services were recognized in the period in which IBM provides the services only when the respective criteria outlined in the accounting rules were met. If those criteria were not met revenue must be recognized over the term of the SO contract in the same manner as the steady state revenue.

Revenue for the IS transition one-time setup services paid by each customer was deferred and recognized over the term of the contract in Web Hosting contracts.

No unbilled revenue accrual was made and the previously recorded unbilled revenue was reversed when the ultimate collectability or receivable was doubtful

Termination charges were not recorded as revenue but as gross income.

Start-up cost were deferred to match against future revenue only if they can be specifically identified as such, are non-recurring, are recoverable under the terms of the contract and are sufficiently covered by termination charges if significant.

Cost incurred in the construction of any facility to be used in system operation contract was accounted for according to the accounting rules for capital assets.

Facilities purchased from potential system operations customers are recorded according to the accounting rules for purchases of customer assets.

References:

<https://w3.ibm.com/w3publisher/accounting-policy-financial-reporting/accounting-practices-interpretations>

ACCOUNTING

AUDIT PROGRAM #10-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Revenue Recognition/Cost Matching

Primary Test: Revenue and Cost Accruals (RR002.P)

Objective:

Verify that revenue and cost accruals are in accordance with generally accepted accounting principles and IBM practices.

Approach:

Review the various accrual types being made with the auditee prior to testing and gain a sound understanding what is being accrued and why. If there are lump sum accruals, make yourself familiar with the calculation methodology.

Select a sample of revenue and cost accruals from the ledger. Focus on quarter-end, year-end activity, and high dollar transactions.

If possible, obtain a listing of all cost and revenue entries and select those revenue accruals for specific contracts / shipments which were not replaced by a billing/final revenue entry or were the billing/final revenue entry is significant different than the accrual. The same applies for cost accruals.

Are the revenue and cost accruals supported by analysis showing how the accrual levels were determined?

If there is a significant difference between the amount accrued and the amount final recorded determine why it is different. Focus on verifying if the accrual level was correctly determined.

Does the accrual method clearly show that only relevant transactions for the period have been included and that none are taken for the future period?

Where revenue accruals are recorded, is there a control process which ensures that the accrual is reversed when the actual revenue is recorded, and tracks the length of time that the accrual remains on the books? Were accruals exceeding a certain time period being reversed/stopped?

Is the accrual methodology consistent from period to period?

Is there any evidence that accrual adjustments are being made without appropriate management approval, or with the intent to significantly alter the corporation's financial results?

Were the accruals based on reasonable and documented evaluations?

ACCOUNTING

AUDIT PROGRAM #10-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Revenue Recognition/Cost Matching

Primary Test: Measurements (RR003.P)

Objective:

Are measurements in place to provide management with visibility of operational efficiency? Are exception conditions identified, reported, and properly handled?

Approach:

Do measurements exist to gauge operational efficiency, process exceptions and identification of areas for corrective action and/or focus?

Is a gross profit analysis being performed which would identify exceptionally high or low gross profit ratios and could indicate problems with cost/revenue recording? Are the exception conditions investigated, documented, and corrected if found in error (country clip levels must be taken into consideration)?

Is there a cost/revenue match process (i.e., are there contracts with costs but no revenue, or revenue with no costs)? Are there exception reports produced to monitor this activity and are revenue or cost accruals being recorded when appropriate?

Is there a measurement in place for aged reconciling items and unreconciled accounts?

Is there management visibility to all measurements and exception activity, are root cause analysis performed and corrective actions taken as appropriate?

ACCOUNTING

AUDIT PROGRAM #10-00-00

ASSESSMENT QUESTIONS

Key Control Question: 3. Revenue Recognition/Cost Matching

Primary Test: Multiple Element Arrangements (RR004.P)

Objective:

To ensure that an effective process exists to identify all Multiple Element Arrangements (MEAs) and to allocate revenue, cost and expenses to the various contracts and/or elements in accordance with generally accepted accounting principles and IBM practices. The main purpose of the accounting principle concerning MEAs is to prevent acceleration of revenue and to ensure that the revenue that is recorded represents fair values of the products or services being sold. Therefore, it applies to elements within multiple element arrangements that include the delivery of one or more element(s) in one quarter and one or more element(s) in a future quarter(s).

Approach:

It is key that the revenue recognition rules relevant for MEAs are understood. Identify the management controls applied to ensure the integrity of revenue allocation for MEAs. This process should include:

The identification of MEAs including the review of contracts with more than one deliverable and the review of multiple contracts entered with the same customer during a given period as separate contract as such arrangements can also qualify as MEAs.

A separability analysis which determines if each element is separately accounted for or not. The analysis should include the determination if the delivered elements have standalone value, objective evidence of fair value exists for the undelivered elements and if there is an explicit or implicit right of return relative to the delivered item. If these criteria are met the elements of a MEA are to be separated into separate units of accounting.

The revenue allocation analysis which first determines if revenue is to be adjusted, at which value it has to be adjusted and then the execution of the adjustment. The analysis should consider if the interest rates for financed items are market rates, the contracted prices for all of the separate elements of the MEA are within the ranges of prices charged when these elements are sold separately and if the MEA contains element(s) that is leased.

Sample testing:

Select a sample of contracts which have already been identified as MEAs. Ensuring coverage of each major revenue category and contract type and verify that revenue was accounted in accordance with the revenue reallocation methodology for MEAs. Note that certain elements and arrangements such as SW elements essential to functionality (e.g., iSeries server sold with OS/400 operation system SW) and Strategic Outsourcing arrangements are not considered as part of or a MEA. Detailed are specified in the respective accounting principles.

Select a sample of potential unidentified MEAs. These could be contractual agreements which contain more than one brand and two or more contractual agreement with one customer signed in a close time frame (If the contract falls into the first half of a month select this month's and the preceding month's contractual agreements. If the contract falls into the second half of a month select this month's contractual agreements). High potentials for MEAs are for example Monthly License Charge (MLC) waivers, Bottom Line Bids and OIOs. Review if the contract was a MEA according to the established accounting rules.

References:

AP REV MEA: <https://w3.ibm.com/w3publisher/accounting-policy-financial-reporting/accounting-practices-interpretations>

ACCOUNTING

AUDIT PROGRAM #10-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Revenue Recognition/Cost Matching

Primary Test: Global Process Services Arrangement (RR005.P)

Objective:

To ensure that an effective process exists to recognize revenue for Global Process Services (GPS) arrangements in accordance with generally accepted accounting principles and IBM practices.

Approach:

It is key that the revenue recognition rules relevant for GPS Arrangements are understood. Identify the management controls applied to ensure the integrity of revenue recognition and cost recoding for GPS Arrangements. This process should include:

Determination whether multiple units of accounting are separable. The separability criteria for Multiple Element Arrangements (MEA) which includes standalone value of the upfront deliverables and existence of fair value for each undelivered element must be met.

Allocation of arrangement consideration to each unit (element) depending on the result of the separability test above. If it is a Single Unit of Accounting (Element) all arrangement consideration for each element should be allocated to the one combined unit of accounting and follow a single revenue recognition method. If it is a Multiple Units of Accounting (Elements) a determination how the overall consideration should be allocated to each phase is required. Arrangement consideration (revenue) should be allocated to each element based on its relative fair value. Relative fair value needs to be based on objective and reliable evidence as defined in the accounting rules for MEAs.

Determination of the appropriate revenue recognition treatment for each unit (element). If services are not separable and deemed to be a single unit of accounting, the entire arrangement would be considered a services arrangement and would result in a deferral of revenue for all elements and recognition over the run phase of the contract. It must be determined if revenue is recognized via the Output Method (The method to measure the extent of services delivered should be a measure of output stated in the contract as the IBM deliverable) or the Straight-line Method (If the arrangement contains multiple deliverables and none of which are considered to be the predominant deliverable or a measure of output is not available or discernable which would be applicable to the contract as a whole, then use of the straight-line method over the term of the contract is appropriate.)

Select a sample of GPS contracts ensuring coverage of major revenue categories and verify if:

The controls and determination processes were accurately performed (e.g., determination of separability) leading to the appropriate revenue recognition decisions.

Contractual clauses that are contingent in nature (e.g. additional payments to IBM for achieving or exceeding specified targets, such as gain-shares and bonuses or payments by IBM to the client if certain contract performance criteria are not achieved, such as loss-shares and penalties) are accounted for according to the accounting practice for contingent revenue.

Deferred start-up costs can be specifically identified in the contract scope, are non-recurring, are realizable and recoverable under the terms of the arrangement, are covered with a termination charge explicit within the contract or stipulate that a customer cannot terminate for convenience, don't exceed the clip level outlined in the accounting guidance, are deferred to match against future revenues and match all other cost recognition criteria outlined in the accounting guidelines.

Cost associated with the transition and transformation that are separable according to the accounting guidelines for MEAs were not deferred as the revenue for these elements was recognized upfront.

ACCOUNTING

AUDIT PROGRAM #10-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Revenue Recognition/Cost Matching

Primary Test: Global Process Services Arrangement (RR005.P)

(Continued)

Only balance sheet items valid for capitalization were capitalized and were in line with the current capitalization limits outlined in the respective accounting guidance.

Cost activities were timely recorded. Verify that there are no significant time lags in the initial stages of a services contract for costs to be received and recorded. If these time lags exist verify if the appropriate cost accruals were be made.

That the total estimated future loss was recorded at the time an estimate of total contract costs indicated that a loss will be incurred on the total contract.

When there was reason to believe that there will be an action that could trigger contract termination, a recoverability analysis of all deferred costs was performed in the light of the circumstances and anticipated collections of termination charges and, if necessary, appropriate write-offs were recorded.

The accounting guidance for services contracts was applied for GPS arrangements that contained only transition and transformation activities but no run phase.

References:

AP REV SO/CLOUD: <https://w3.ibm.com/w3publisher/accounting-policy-financial-reporting/accounting-practices-interpretations>

ACCOUNTING

AUDIT PROGRAM #10-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Revenue Recognition/Cost Matching

Primary Test: BOP Controls (RR006.P)

Objective:

To evaluate accounting controls over the recording of hardware cost, inventory data and depreciation in FIW.

Approach:

Review the applied measurement deflation methodology in the country and if process controls and key measurements were established to monitor the deflation process and to ensure correct deflation in FIW. Review the process for deflation exclusions and manual deflation adjustments.

Select a sample of product numbers and test, if the deflation in FIW was performed correctly. Focus on product divisions where rate changes have occurred.

Select manual deflation adjustments (Voucher Disbursement Entries (VDEs)) for accounting and check for appropriate supporting documentation and appropriate management approval.

Request the balances of control accounts for operating lease for the last three months (including year-end) and test if deviations were properly investigated and deflation adjustments were timely performed.

Review the process controls and the measurements to verify if deflation in marketing countries is in line with inflation in plants.

ACCOUNTING AUDIT PROGRAM #10-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Revenue Recognition/Cost Matching

Optional Test: Labor and/or employee concur expense costs

Objective:

To ensure that employee costs were recognized in accordance with generally accepted accounting principles and IBM practices.

Missing or late labor and/or employee concur expense information items may not be aligned between the BMS and ledger interface. The financial records (costs) will not be booked accordingly. If the items are not cleared in the interface, financial reporting of those related account IDs will be incorrect, resulting in incorrect cost, too high profitability, and incorrect incentives due to employees.

Approach:

This is a self-serve evaluation performed by the Technology team, possible to be ran based on country/brand.

For more accurate information, include the contract number, account ID, and the organization (brand) when submitting the data analysis request to the Technology Team. This will allow better alignment to the database where the records are stored.

The information can be requested during IDR analysis and could provide valuable information for contract sampling for testing.

ACCOUNTING AUDIT PROGRAM #10-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. I&E Account Analysis and Reporting

Objective:

Ensure there is a control process at unit level to review trends or fluctuations in Income & Expense accounts by reviewing actual performance both against plan and against prior month actual performance.

Ensure all I&E accounts have a designated owner with responsibility for reviewing performance analysis.

Risk/Exposure:

Any account for which ownership has not been formally assigned and accepted may not be reviewed with the necessary frequency and diligence.

In addition, failure to review trends and performance against plan could allow account(s) to contain erroneous or improper information, thereby distorting results and/or concealing irregularities.

ACCOUNTING

AUDIT PROGRAM #10-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. I&E Account Analysis and Reporting

Primary Test: Material Trend Account Analysis (IE001.P)

Objective:

Investigate material trends or fluctuations in I&E accounts and compare with records of reviews to ensure management awareness and focus on the integrity of the financial data.

Approach:

The auditor should review the trend analysis work performed as part of the audit planning activity and supplement that analysis with the following tests:

- Ensure all accounts have been allocated an owner with responsibility for performance of analysis.
- Review significant account movements and ensure that the trends are identified, reviewed and actioned.
- Select a judgmental sample of I&E accounts for review, either by reviewing month end reporting for successive months (for a three-month period) or by use of data analysis techniques. The sample selection should include accounts with the following characteristics:
 - Unusual trend
 - Deviation from planned level
 - High volume accounts, level of amounts
 - Peaks, static volumes
 - Unexpected credits/debits, including account balance contradictory to account type
 - New/closed balances
 - Sensitive accounts (write-offs; accruals; 'miscellaneous' and suspense accounts)
- Compare the sample with management reporting to confirm the adequacy of management review.

ACCOUNTING

AUDIT PROGRAM #10-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. I&E Account Analysis and Reporting

Secondary Test: Currency Translation (IE002.S)

Objective:

- Ensure that all accounts were treated correctly for currency conversion, that the appropriate currency exchange rates were applied throughout the currency translation process, and that all manual adjustments to the translation gain/loss accounts were appropriate, authorized and documented.
- Ensure the quarterly reconciliation of the equity translation adjustment (ETA) account is performed and approved on a timely basis and in conjunction with IBM accounting procedures.

Approach:

For a period of three months:

- Request a copy of the conversion rate table to compare the exchange rates to those advised by the headquarters accounting function.
- Test manual movements into the exchange gains/losses (translations) accounts to ensure their justification, documentation, and authorization.
- Request and review the ETA reconciliations for a six-month period to ensure accuracy, timeliness and approval by the appropriate level of management.
 - Reconciliations are being performed and approved as required.
 - Ensure the monthly net addition or deduction to the equity translation adjustment is being proven before the figures are submitted.
 - Test a sample of the calculations of some of the differences to ensure it supports the figures provided by management.
 - Ensure all figures in the calculations agree to the ledger and figures used elsewhere in reporting.
 - All reconciling items are fully documented, justifiable and minimal.

Worldwide Consolidation Accounting Monthly Currency Exchange Rates:

<https://w3.ibm.com/w3publisher/currency/monthly-exchange-rates>

ACCOUNTING AUDIT PROGRAM #10-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 5. Business Continuity Planning

Objective:

To determine if process owners have developed, reviewed, and tested business continuity plans (BCP) so that in the event of natural or man-made disasters, pre-defined plans can be executed to maintain business operations.

Risk/Exposure:

Any material disruption of business operations due to natural or man-made disasters can lead to significant financial loss not only IBM but potentially also for IBM's customers. In the accounting process it could also lead to an inability to ensure that transactions are promptly accounted for. Many business processes have been centralized, and often operations are conducted in locations which may be susceptible to business disruption. As a result, it is important that business and process owners have developed plans to maintain continuity of business operations in those centralized operations.

ACCOUNTING

AUDIT PROGRAM #10-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 5. Business Continuity Planning

Secondary Test: Business Continuity Planning (BC001.S)

Note: Business continuity risks become increasingly important in the environment where there are centers of excellence and multiple countries are supported by one center. This should therefore be an important test and considered primary in that environment. IT testing for disaster recovery is handled by the IT audit teams.

You should refer to BPBC127 Guideline for Business Process - Area Business Continuity Planning in the reference section of this audit program, in support of your review.

Objective:

To determine if process owners have developed, reviewed, and tested business continuity plans (BCP) so that in the event of natural or man-made disasters, pre-defined plans can be executed to maintain business operations.

Approach:

A guideline has been put in place for process owners providing guidance on business continuity planning (BCP). It should be noted by the audit team that this is a guideline rather than a specific instruction. However, it will act as a guide for you to understand whether key risks have adequately been considered for BCP and enable you to make an informed judgement regarding the risk mitigation in place. Different processes will have their own unique risk profiles regarding the probability and impact of an interruption to business operations. For example, the probability and impact of a business interruption for a process which has been centralized for multiple countries in a location with less-developed infrastructure is likely greater than that of a process supporting only a single country in location with robust infrastructure. As such, auditor judgment will be required to determine how important BCP is for a particular process, as well as to determine the reasonableness of any plans that have been designed by the process owners.

In the initial data request information ask for the process owner to confirm what plans have been implemented to ensure business continuity in the event of a natural or man-made disaster. Review that non-I/T process elements have been adequately considered in the BCP.

The following list of questions has been designed based on the CIO 127 Guideline for Business Process - Area Business Continuity Planning:

1. Has management assessed the risk to business continuity (BC) and identified the key business processes and activities that require formal business continuity planning based on impact to the business?
 - Identification and prioritization of business processes' activities in addition to prioritization done through by the Enterprise Process Owner (level 3 plans)
 - Understanding and documentation of the potential impact (financial, operational and reputation) to IBM from business disruptions from uncontrolled, non-specific events
 - Identification of the legal and regulatory requirements for the institution's business functions and processes
 - Review and approval from senior management

ACCOUNTING

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 5. Business Continuity Planning

Secondary Test: Business Continuity Planning (BC001.S)

(Continued)

2. For critical business areas and processes, has the maximum acceptable outage (MAO) been established and considered in the BC plan?
 - Estimation of maximum allowable outage (MAO), based on the “acceptable level” of losses, associated with the institution's business functions and processes
 - Estimation of recovery time objectives (RTOs), recovery point objectives (RPOs), and Maximum Tolerable Downtime (MTD) of applications and supporting technology required for business operations
 - If applicable, recommended critical path recovery order
 - Reconciled with agreed IT recovery priorities (Business Criticality Values, RTO/MTD)
3. For business areas and processes where there is a business continuity risk, has the BC Plan been created, reviewed, and approved by management?
 - The organization shall establish documented procedures for responding to a disruptive incident and how it will continue or recover its activities within a predetermined timeframe.
 - Top management shall review the organization's BCMS, at planned intervals, to ensure its continuing suitability, adequacy and effectiveness.
4. Does the Plan document the specific continuity and recovery actions for all four impact scenarios (workforce, workplace, IT, and regional) that will need to take place to mitigate risk to the business if the impact lasts longer than the documented business' maximum acceptable outage?
 - Flexible to respond to unanticipated threat scenarios and changing internal conditions
 - Focused on the impact of various threats that could potentially disrupt operations rather than on specific events
 - Effective in minimizing service disruptions and financial loss through the implementation of mitigation strategies
5. Does the BC plan clearly identify dependencies such as site resources, IT, suppliers, process interdependencies, partners, non-IT services and been reviewed and synchronized with the appropriate crisis management team?
 - An effective BCP coordinates across its many components, identifies potential process or system dependencies, and mitigates the risks from interdependencies.
6. Is there an established, management-endorsed business continuity team to oversee the maintenance, invocation, and execution of the BC plan?
 - Defined roles and responsibilities for people and teams having authority throughout an incident
 - Additional teams as necessary to facilitate recovery execution per the chosen strategy

ACCOUNTING

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 5. Business Continuity Planning

Secondary Test: Business Continuity Planning (BC001.S)

(Continued)

7. Are personnel aware and educated on the plan, strategy, and roles?
 - Business continuity training should be an integral part of the BCM
 - Comprehensive training program should be developed for all employees, conducted at least annually, and kept up to date to ensure that everyone understands their current role in the overall recovery process
 - An audit trail should be maintained to document management's training efforts
8. Is there a regularly executed process for validating the Plan through tests including key elements such as people, technology (data, systems, applications, and telecommunications), and locations that house staff and technology to ensure fit for purpose and alignment with current risks?
 - Tests should be based on appropriate scenarios that are well planned with clearly defined aims and objectives that promote continual improvement
 - Produce formalized post-exercise reports that contain outcomes, recommendations and actions to implement improvements and are reviewed with management
 - Tests are conducted at planned intervals and when there are significant changes within the organization or to the environment in which it operates
9. FOR COMMERCIAL CONTRACTS: Where IBM has customer contracts including business continuity planning, have BC plans been implemented in accordance with contractual obligations?

FOR CUSTOMERS WITH REGULATORY REQUIREMENTS: Does the plan accommodate and account for the any unique rigor or accommodations for the regulatory body.

- Adherence with contractual obligations contained in the IBM Responsibilities section of the Schedule or similar agreement. Including communication, recovery parameters, specialized technical designs.
- Special considerations to meet unique regulatory requirements (HIPAA – US Healthcare has unique wording on backup, protection, and availability of data).

Reference:

AI CTRL 14-001 Business Continuity Planning:

<https://w3.ibm.com/w3publisher/worldwide-accounting-controls/accounting-instructions>